

56. What is a tuple in Python?

Answer:

A tuple is a collection of items that is **ordered and cannot be changed (immutable)** after creation.

Example:

```
id="a2k8ps"
```

```
numbers = (10, 20, 30)
```

```
print(numbers)
```

Output:

```
(10, 20, 30)
```

Features of tuples:

- Ordered
- Immutable (cannot modify)
- Allows duplicate values
- Can store different data types

57. What is the difference between a list and a tuple?

Answer:

List	Tuple
Mutable (can change)	Immutable (cannot change)
Uses square brackets []	Uses parentheses ()
Slower than tuples	Faster than lists
More memory usage	Less memory usage

Example:

List:

```
id="m0v3qf"
```

```
a = [1, 2, 3]
```

```
a[0] = 10
```

Tuple:

```
id="x2k9fd"
```

```
b = (1, 2, 3)
```

58. What is a dictionary in Python?

Answer:

A dictionary is a collection of data stored in **key-value pairs**.

Example:

```
id="p6x8nz"
```

```
student = {  
    "name": "John",  
    "age": 20  
}
```

```
print(student["name"])
```

Output:

John

Features:

- Stores data as key-value pairs
- Keys must be unique
- Mutable
- Fast data access

59. How do you add, update, and delete elements in a dictionary?

Answer:

Add element:

```
id="h3q0mz"
```

```
student["city"] = "Hyderabad"
```

Update element:

```
id="y7k1la"
```

```
student["age"] = 21
```

Delete element:

```
id="v8p4ws"
```

```
del student["age"]
```

Example:

```
id="j4x6cf"
```

```
student = {"name": "John"}
```

```
student["age"] = 20
```

```
print(student)
```

Output:

```
{'name': 'John', 'age': 20}
```

60. What is the difference between keys() and values() in a dictionary?

Answer:

keys():

- Returns all keys from a dictionary.

values():

- Returns all values from a dictionary.

Example:

```
id="r2m5ks"
```

```
student = {  
    "name": "John",  
    "age": 20
```

```
}
```

```
print(student.keys())
```

```
print(student.values())
```

Output:

```
dict_keys(['name', 'age'])
```

```
dict_values(['John', 20])
```